Jacek Karwowski

jacek.karwowski@cs.ox.ac.uk Academic website

EDUCATION

Ph.D. in Computer Science, University of Oxford *2022 - ongoing*
Specialization: programming languages, AI safety, information geometry

M.Sc. in Mathematics, University of Warsaw *2018 - 2021*
Thesis: Formal semantics of a reversible language

B.Sc. in Computer Science, University of Warsaw *2018 - 2021*
Thesis: Low-latency Apache Parquet library (blogpost)

B.Sc. in Mathematics, University of Warsaw *2015 - 2018*
Thesis: Products in positive opetopic sets

PUBLICATIONS

J. Karwowski, F. Nielsen, and S. Staton. Compositional Information Geometry of Probabilistic Programs, Oct. 2026. https://arxiv.org/abs/2610.17886

J. Karwowski, Y. Kaddar, Z. Ye, E. S. Whitammer, and S. Staton. Likelihood hacking in probabilistic program synthesis. In E. Perković and D. Malinsky, editors, *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence*, volume 337 of *Proceedings of Machine Learning Research*, pages 2744–2791. PMLR, Aug. 2026

J. Karwowski and F. Nielsen. Geometric structures and deviations on James' symmetric positive-definite matrix bicone domain, Mar. 2026. https://arxiv.org/abs/2603.02483

J. Karwowski and R. Douglas. Incoherence in goal-conditioned autoregressive models. In E. Khan, Y. Li, A. Solin, and A. Ramdas, editors, *Proceedings of the 29th International Conference on Artificial Intelligence and Statistics*, volume 300 of *Proceedings of Machine Learning Research*, pages 2611–2619. PMLR, May 2026

J. Karwowski and F. Nielsen. Hilbert geometry of the symmetric positive-definite bicone. In F. Acosta, S. Azeglio, B. Tolooshams, C. van de Geijn, C. Shewmake, S. Sanborn, and N. Miolane, editors, *Proceedings of the 4th (2025) and 3rd (2024) NeurIPS Workshops on Symmetry and Geometry in Neural Representations*, volume 282 of *Proceedings of Machine Learning Research*, pages 263–281. PMLR, 14 Dec 2024–07 Dec 2025 2026

J. Janiak, J. Karwowski, C. S. Mangat, G. Giglemiani, N. Petrova, and S. Heimersheim. Characterizing stable regions in the residual stream of LLMs. In *NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning*, Nov. 2024

R. Douglas, J. Karwowski, C. Bae, A. Draguns, and V. Krakovna. Limitations of agents simulated by predictive models. In *ICLR 2024 Workshop on Large Language Model (LLM) Agents*, 2024

J. Karwowski, O. Hayman, X. Bai, K. Kiendlhofer, C. Griffin, and J. M. V. Skalse. Goodhart's law in reinforcement learning. In *The Twelfth International Conference on Learning Representations, ICLR*, 2024

N. Ackerman, C. E. Freer, Y. Kaddar, J. Karwowski, S. Moss, D. Roy, S. Staton, and H. Yang. Probabilistic programming interfaces for random graphs: Markov categories, graphons, and nominal sets. *Proc. ACM Program. Lang.*, 8(POPL), Jan. 2024

V. Choudhury, J. Karwowski, and A. Sabry. Symmetries in reversible programming: From symmetric rig groupoids to reversible programming languages. *Proc. ACM Program. Lang.*, 6(POPL), Jan. 2022

TEACHING & ACADEMIC

Teaching

Teaching Assistant - Probability Theory, University of Oxford 2024
Teaching Assistant - Introduction to AI Alignment, Oxford AI Safety 2024
Teaching Assistant - Bayesian Statistical Probabilistic Programming, University of Oxford 2024
Teaching Assistant - Lambda Calculus and Types, University of Oxford 2023
Grader - Linear Algebra, University of Warsaw 2021

Academic Service

Reviewer - AISTATS 2026, NeurIPS 2025, ICLR 2025, NeurIPS 2024 (top reviewer), CAV 2023
Program Officer - PL in ML Conference 2017

Scholarships & Fellowships

Research Scholarship - SERI ML Alignment & Theory Scholars 2023
PhD Scholarship - Improving the Long-Term Future, Open Philanthropy 2022 - 2025
Fellowship - Global Talent Attraction Program, Indiana University 2019

WORK EXPERIENCE

Researcher - University of Oxford 2025 - ongoing

I'm working on an Opportunity Seed project "SynthStats: GFlowNet-steered probabilistic program synthesis for safer AI", funded by UK's Advanced Research and Invention Agency.

Researcher - National Institute of Informatics 2025

I worked on the information-geometric aspects of probabilistic programming languages theory at Sugiyama Lab, National Institute of Informatics in Tokyo, Japan.

Research lead - Supervised Program for Alignment Research 2025

I supervised the project "Understanding and verifying the autoregressive conditioning hypothesis". The team consisted of six people with backgrounds in computer science and mathematics.

Researcher - Stanford Existential Risks Initiative 2023

As part of SERI ML Alignment Theory Scholars program, I worked with Victoria Krakovna and others on understanding how can powerseeking arise in predictive models.

Researcher - Oxford AI Safety Labs 2022 - 2023

I worked with Joar Skalske at the Future of Humanity Institute and others on understanding Goodhart's law in reinforcement learning.

Quantitative Developer - Xantium 2021 - 2022

Xantium is the algorithmic trading division of Tudor Corporation. I worked on volatility trading, where I built models, implemented execution and managed datasets. Work details under NDA.

SWE Intern - G-Research 2020

G-Research is a quantitative research company, developing algorithms for predicting movements in financial markets. I worked on programmers tools for the simulations platform. Work details under NDA.

Research Intern - Indiana University 2019

As part of Global Talent Attraction Program, I conducted research on understanding the semantics of reversible programming languages, in the context of Homotopy Type Theory.

SWE Intern - Microsoft 2019

I worked on Office365 back-end, adding new features and optimizing the existing ones, and deploying code to production. Work details under NDA.

Research Intern - Golem Factory GmbH 2017 - 2018

Golem is a blockchain startup, creating a platform for sharing computational resources on Ethereum. I developed a novel probabilistic algorithm for verification of results in an untrusted environment (whitepaper, blogpost), and implemented it for distributed hyper-parameter optimization (blogpost).

Data Scientist - Applica.ai 2016 - 2017

Applica is a startup working on ML tools for NLP. I was responsible for creating new models: from designing experiments to prototyping models in Python and implementing them on the production cluster.

SWE Intern - University College London 2014 - 2015

Over the academic year, I have worked on developing a platform for testing high-frequency trading algorithms, and on an open-source medical data platform.